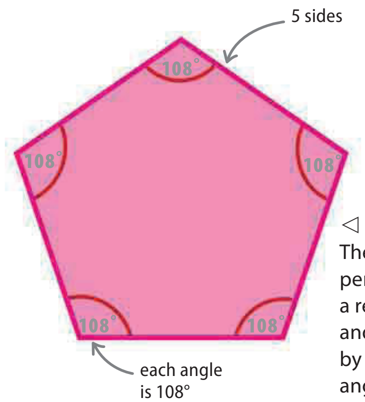
A formula for the interior angle sum

The number of triangles a convex polygon can be split up into is always 2 fewer than the number of its sides. This means that a formula can be used to find the sum of the interior angles of any convex polygon.

Sum of interior angles = $(n - 2) \times 180^\circ$

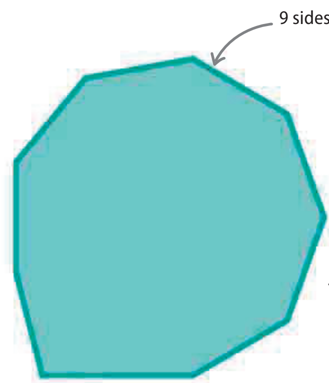


number of sides

$$(5 - 2) \times 180^\circ = 540^\circ$$

sum of its interior angles

◁ **Regular pentagon**
The interior angles of a regular pentagon add up to 540°. Because a regular polygon has equal angles and sides, each angle can be found by dividing by the number of angles: 540° ÷ 5 = 108°.



number of sides

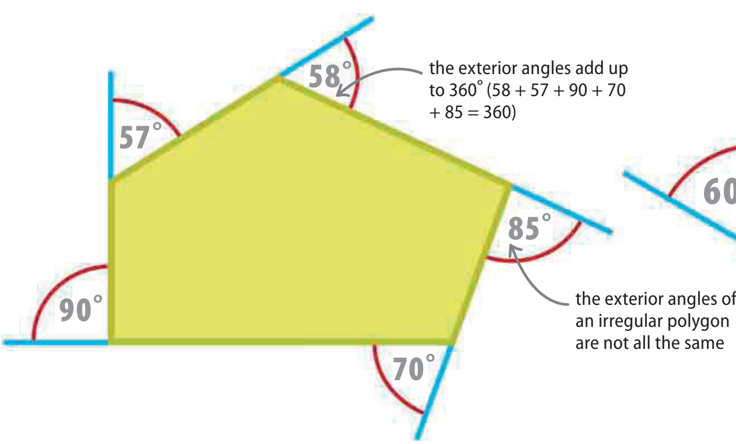
$$(9 - 2) \times 180^\circ = 1{,}260^\circ$$

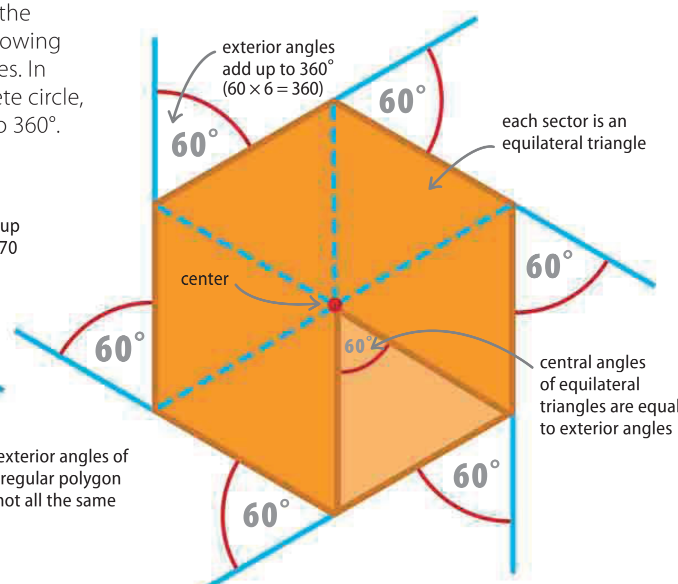
sum of its interior angles

◁ **Irregular nonagon**
The interior angles of an irregular nonagon (9 sides) add up to 1,260°. Because the angles are different sizes, individual angles cannot be found from the sum of the interior angles.

Sum of exterior angles of a polygon

Imagine walking along the exterior of a polygon. Start at one vertex, and facing the next, walk toward it. At the next vertex, turn the number of degrees of the exterior angle until facing the following vertex, and repeat until you have been around all the vertices. In walking around the polygon, you will have turned a complete circle, or 360°. The exterior angles of any polygon always add up to 360°.





△ **Irregular pentagon**
The exterior angles of a polygon, regardless of whether it is regular or irregular add up to 360°. Another way to think about this is that, added together, the exterior angles of a polygon would form a complete circle.

△ **Regular hexagon**
The size of the exterior angles of a regular polygon can be found by dividing 360° by the number of sides the polygon has. A regular hexagon's central angles (formed by splitting the shape into 6 equilateral triangles) are the same as the exterior angles.